



Exercise Sheet 01

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Due: May 12, 2025

Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a jupyter notebook.

1. Set theory

(a) Answer the following questions:

1P

- What's the power set of $\{x, y, z\}$?
- What is the intuitive explanation of the power set of the natural numbers $\mathcal{P}(\mathbb{N})$?
- Give an example of a relation X that is not reflexive, i.e. $(a, b) \in X$ but $(b, a) \notin X$?
- Prove that $X \cup (\emptyset \cap Y) = X$

(b) Show that the following statements are equivalent:

2P

- $A \subseteq B$
- $A \cap B = A$
- $A \cup B = B$

2. Relations

Prove the following:

(a) A relation is called equivalence relation if it is reflexive, symmetric and transitive.

2P

Let R be a relation on a group G , where xRy for $x, y \in G$ if and only if there exists $g \in G$ such that $y = g^{-1}xg$.

(A group G is a non-empty set of elements together with a binary operation $*$: $G \times G \rightarrow G$ with the following properties:

- There exists neutral element $1 \in G$ with $g1 = g$ and $1g = g$ for all $g \in G$ and $1^{-1} = 1$.
- For all $g \in G$ there exists $g^{-1} \in G$ such that $gg^{-1} = g^{-1}g = 1$.
- G is associative, i.e. $(gh)i = g(hi)$ for all $g, h, i \in G$.

Here we wrote gh instead of $g * h$.)

(b) Prove that an equivalence relation which is also a function is necessarily the identity function?

1P

The identity $I : A \rightarrow A$ is defined by $I(a) = a$ for all $a \in A$.

3. Formal Languages and Automata

(a) Write a context-free grammar for the following languages:

2P

- $L_1 = \{a^n b^m \mid m \geq n, n > 0\}$
- $L_2 = \{a^l b^k c^{l+k} \mid l, k \in \mathbb{N}\}$



(b) Given the grammar $G = (\{S, A\}, \{a, b\}, P, S)$ with

2P

$$P = \{S \rightarrow aAS$$

$$A \rightarrow SbA$$

$$A \rightarrow SS$$

$$S \rightarrow a$$

$$A \rightarrow ba\}$$

(i) What production rules can we use to derive $w = aabbaa$?